

35 Polymerisation

A • 2026 examination • Concept guide and original exam-style practice

Understand the concept

Condensation polyamides and polyesters are vulnerable to hydrolysis because their links can be cleaved; addition polymers with C-C backbones resist hydrolysis. A repeat unit should show bonds that extend to adjacent units, and monomers can be deduced by cutting linkage bonds and restoring their functional groups.

Worked example: follow the reasoning

Polyester repeat $\text{-O-CH}_2\text{-CH}_2\text{-O-C(=O)-C(=O)-}$ corresponds to a diol and dicarboxylic acid.

Quick check • multiple choice

A Which polymer backbone lacks hydrolysable ester or amide links?

[1]

- A Polyester
- B Polyamide
- C Poly(ethene)
- D Protein

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why a polyamide may be hydrolysed by aqueous acid but poly(ethene) is relatively resistant.

[3]

Answers and explanation

Quick check: C. Poly(ethene)

Question 1 [3 marks]

Amide C-N links undergo hydrolysis [1]; acid catalyses cleavage [1]; poly(ethene) C-C backbone lacks such hydrolysable functional groups [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.